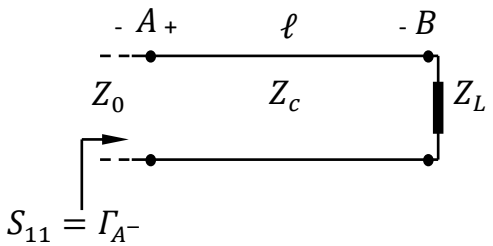
Lab. 1 – Numerical analysis of matching circuits

Assigned: **Friday March 13th, 2026**
Due: **Friday March 24th, 2026 at class starting**

Problem nr. 1

a) Design a lossless Quarter Wave Transformer (QWT), consisting of a transmission line section with length $\ell = \lambda_0/4$ at the (design) center-band frequency f_0 and characteristic impedance $Z_C = \sqrt{Z_0 Z_L}$, that matches a load with impedance Z_L to the characteristic impedance Z_0.



Evaluate Z_C, for the following two cases:

1. $Z_0 = 50\ \Omega$, $Z_L = 100\ \Omega$;
2. $Z_0 = 50\ \Omega$, $Z_L = 200\ \Omega$.

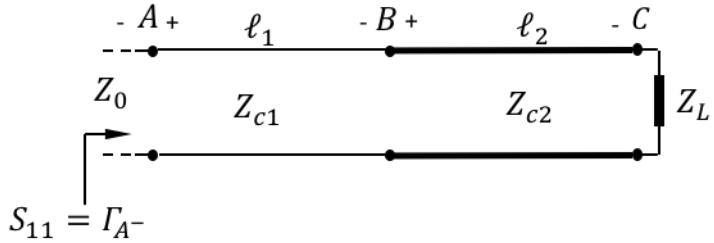
b) Using Matlab™, write a script to compute the frequency behavior of $|S_{11}(f)|$ set in the range $[0,2f_0]$, and plot it as a function of f/f_0, for the two different cases.

c) Determine the frequency interval $[f_1, f_2]$ in which $|S_{11}(f)|_{dB} < -20$ dB, and compute the fractional bandwidth $(f_2 - f_1)/f_0$.

Problem nr. 2

Referring to the case 1 of Problem 1 ($Z_0 = 50\ \Omega$, $Z_L = 100\ \Omega$), modify the matching circuit to enlarge its fractional bandwidth.

In view of this, consider a 2-port network consisting in the cascade of two sections of line with characteristic impedance $Z_{C1} = 60\ \Omega$ and $Z_{C2} = 84\ \Omega$, respectively. Each line section has length $\ell_i = \lambda_i/4$, with $i = 1, 2$ where λ_i is the wavelength in each line section evaluated at the design frequency f_0.



a) Using Matlab™, write a script to compute the frequency behavior of $|S_{11}(f)|$ set in the range $[0,2f_0]$, and plot it as a function of f/f_0.

(Hint: you can directly evaluate $S_{11}(f)$ or either represent each line section with its *ABCD* matrix, then compute the *ABCD* matrix of the entire network and finally evaluate S_{11}).

b) Determine the frequency interval $[f_1, f_2]$ in which $|S_{11}(f)|_{dB} < -20$ dB, and compute the fractional bandwidth $(f_2 - f_1)/f_0$. Compare with the corresponding case in Problem 1.